

Key Words: Placebo, double-blind trial, toxicity, dose, efficacy

Starter: What drugs do you know both illegal and legal?

EXT: Why should we test drugs before we used them?



History of Thalidomide



**Chemie Grünenthal
Pharmaceutical
Company** in W. Germany
Developed the
Drug in 1950s

Testing showed it was
impossible to overdose

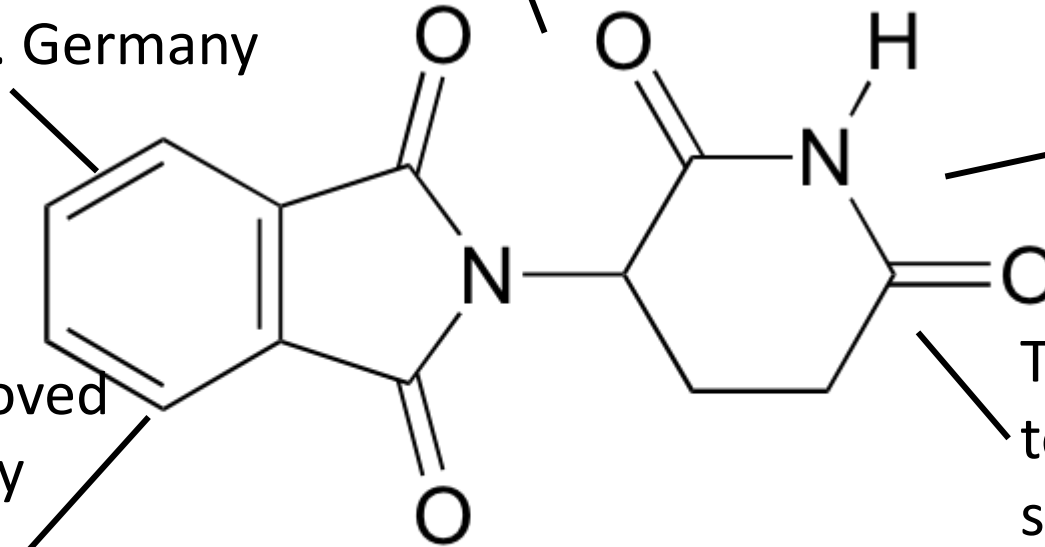


Animal testing was not
performed in pregnant animals



Primarily
prescribed as a
Sedative and
hypnotic

1962 it was proved
to be caused by
the drug and
was banned.



Then it was used
to stop 'Morning
sickness' and
'Nausea' in
pregnant women

Then 5,000
-7,000
babies were
Born with
Phocomelia



*1967 showed the
drug could help
leprosy symptoms*



History



History



Thalidomide: 10,000 cases were recorded globally with only 50% of these children surviving.



Key Words: Placebo, double-blind trial, toxicity, dose, efficacy

Starter: Answer these three review questions below.

1. Why did Alexander Fleming name the antibiotic penicillin? [1 marks]



2. Foxgloves are the source for _____ and _____ trees are the source of _____ [3]

3. Describe how penicillin was discovered [4]

Review

1. Why did Alexander Fleming name the antibiotic penicillin?
[1 marks]

Because the antibiotic came from the mould called penicillium [1]

2. Foxgloves are the source for digitalis and willow trees are the source of aspirin. [3]

3. Describe how penicillin was discovered [4]

- ☐ Alexander Fleming grew bacteria on agar plates and went on holiday [1]
- ☐ He left the lid open on his plates and came back and found mould growing [1]
- ☐ Mould killed bacteria, he tried to extract active juice and was unsuccessful [1]
- ☐ Florey and Chain were successful and Pfizer mass produced the penicillin. [1]

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Starter: Let's do a quiz on what we learnt last lesson. 1□ 10

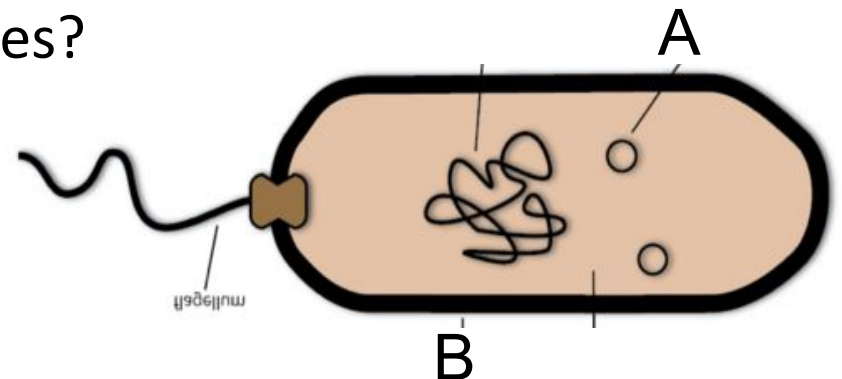
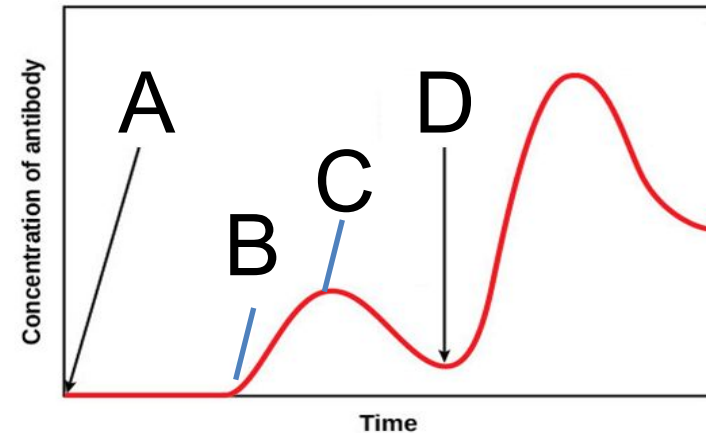
TIME FOR A

QUIZ!



Quiz:

1. What drug comes from foxgloves?
2. Which drug comes from willow trees?
3. Why did mould grow on Alexander Flemings agar plates?
4. Most new drugs are developed by _____ working for _____ companies.
5. At what stage on the graph will there be memory cells? **A | B | C | D**
6. At what point on the graph do the immune responses occur?
7. A vaccine contains a _____ form of a pathogen that we inject.
8. Why do antibiotics not work on viruses?
9. Name two bacterial diseases.
10. What is part A in the bacterial cell?



LOb: Understand how new drugs are developed and tested.

Keywords: Placebo, double-blind trial, toxic dose, efficacy

Learning Outcomes:

- Describe what makes a good medicine
- Explain the steps involved with modern drug trials
- Explain double-blind and placebo

Did you know... there are always new drug trials happening but not everyone wants to try and test an untested drug. In the future you could choose to be part of a drug trial and earn as much as £2000. But would you want to take the risk?



What makes a good medicine?

Effective: Cure,
prevent improve
health

Safe: Not too toxic or
have unacceptable
side effects

Stable: Used
under normal
circumstances
and can be
stored

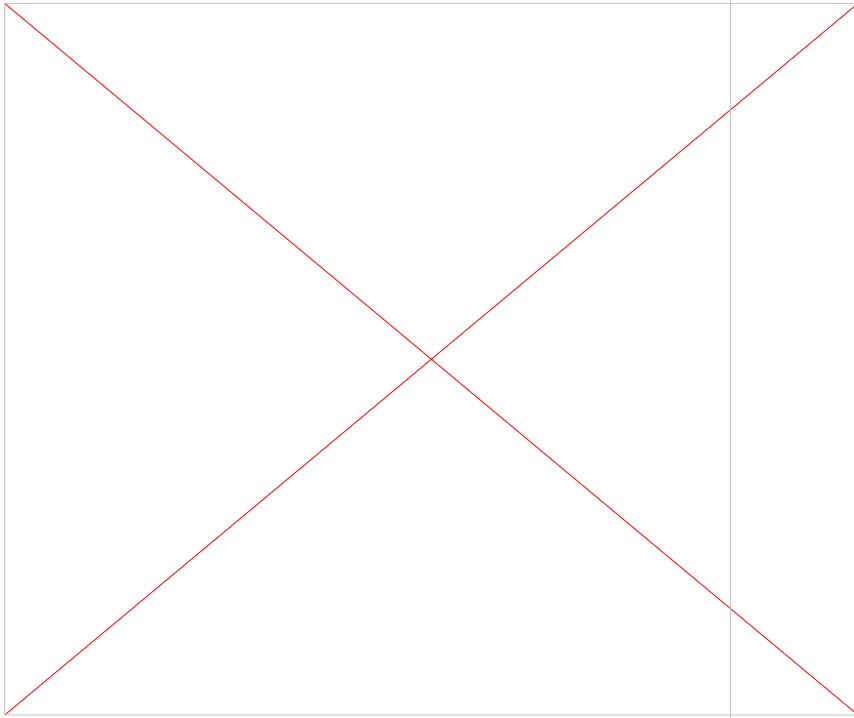


**Successfully
removed:**
Able to be
removed after
it's job is done

All drugs are extensively test for these three things...

Dose

Concentration
*Concentration of drug to be
used and how often.*

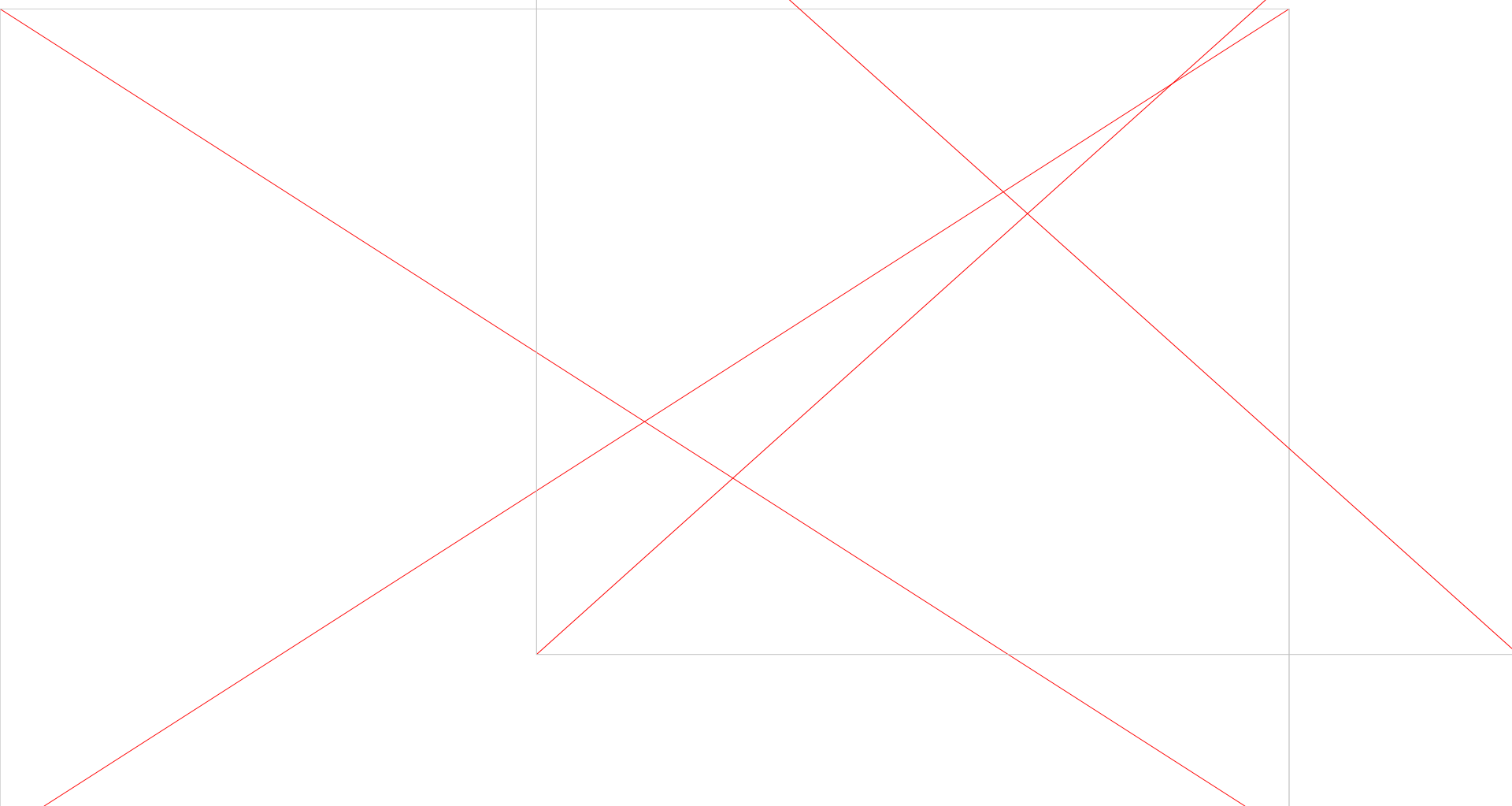


Three key words you need to know.

Effic

acy

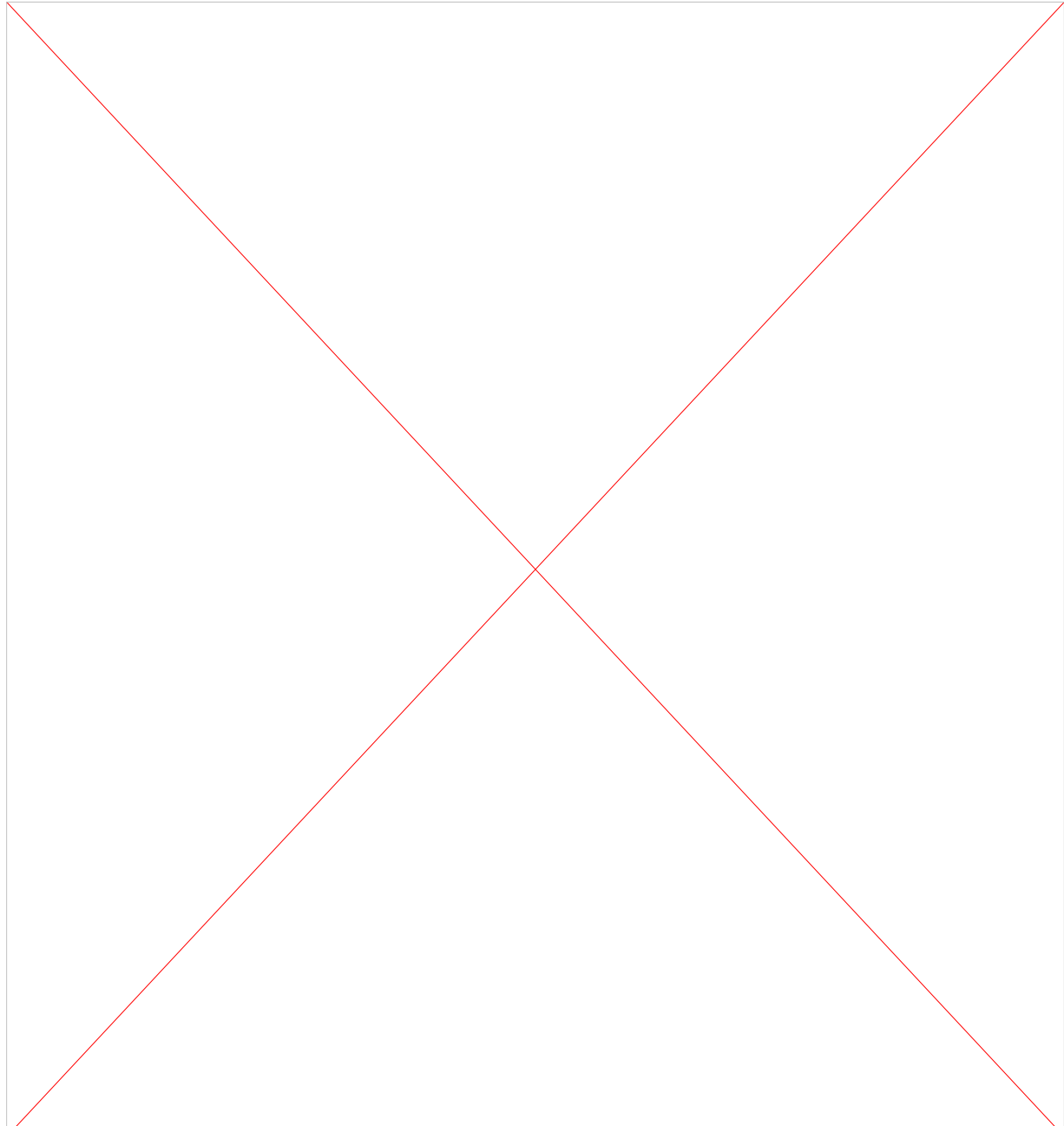
*How effective the drug
is at treating disease.*



Three key words you need to know.

Toxic

ity
*Does the drug cause
side effects that make
a patient ill.*



Three key words you need to know.

**Dosa
ge
Efficacy
Toxicity**

Side effects making the person
ill

The concentration of the
drug
to be used and how often it
is given.

Whether the drug works
to
treat the disease

LOb: Understand how new drugs are developed and tested.

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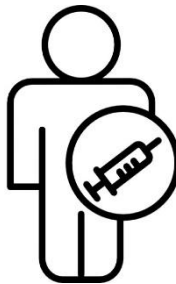


Developing Drugs.

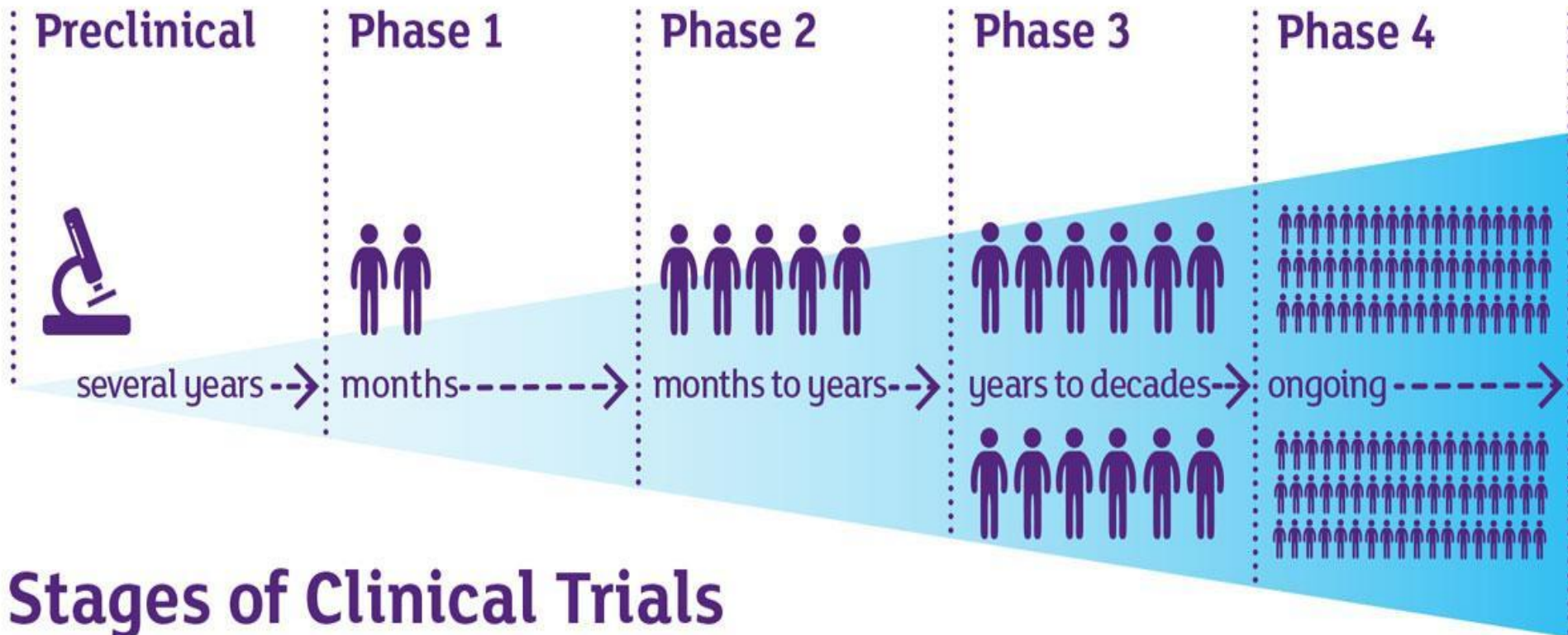
On average it takes 10 -12 years and £550+ million to develop a new drug.

New drugs go through a rigorous round of testing.

Dosage
Toxicity
Efficacy



Activity: Watch the video explaining the stages of modern drug trials. Complete your sheet with the steps.



1. Researchers target a disease and develop ideas for treatment. Drugs are made in the lab. Clinical trials are used to establish if the drug is:

✓ Safe

✓ Effective

2 Pre-Clinical Trial Stage 0.

Try to establish the Dose ; which means the amount of drug we need to give.

3. Stage 1. Who are the people the drugs are first tested on and why?



Young men

Because you don't know if the women are pregnant and you can't take risks

4. Stage 2 of clinical trials.

Use patients that are ill

It needs lots of Money

Only the best Universities



5. Stage 3 clinical trials involves a lot more



money



time



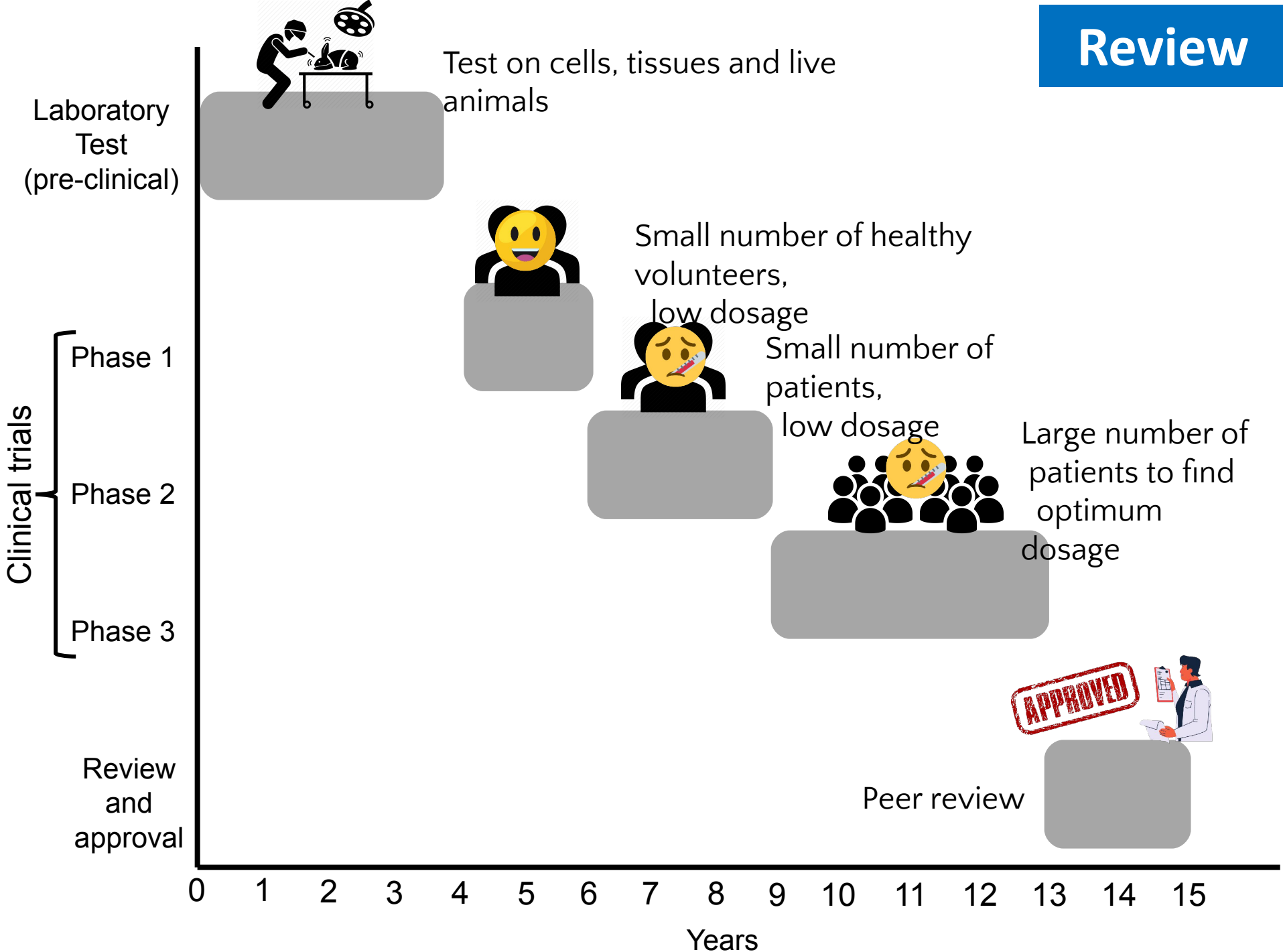
More patients



6. Stage 4. **Describe this stage**

To find new uses in new patient groups and find new
Combinations and doses. Can take up to 8 years.

Look at the study report submitted to the regulatory authority.



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Learning Outcomes:

✓ Describe what makes a good medicine

✓ Explain the steps involved with modern drug trials

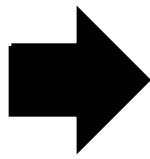
□ Explain double-blind and placebo

Did you know... there are always new drug trials happening but not everyone wants to try and test an untested drug. In the future you could choose to be part of a drug trial and earn as much as £2000. But would you want to take the risk?

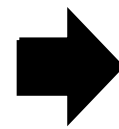


What happens in a good clinical trial?

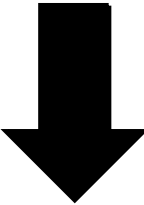
Clinical trials involving real patients



Patients are put into **groups** and have a **doctor**



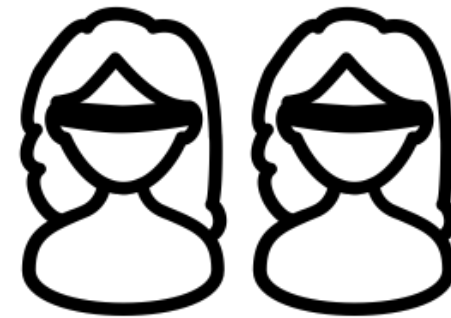
One group receives a **fake drug** made of **sugar** called a **placebo**. Other group takes the real drug



Results are **published in medical journals**. The data is **peer reviewed** by other scientists who are working in the same area. This **prevents false claims**.



Doctor doesn't and patients don't know which drug is being taken. This is called **Double Blind Trial**

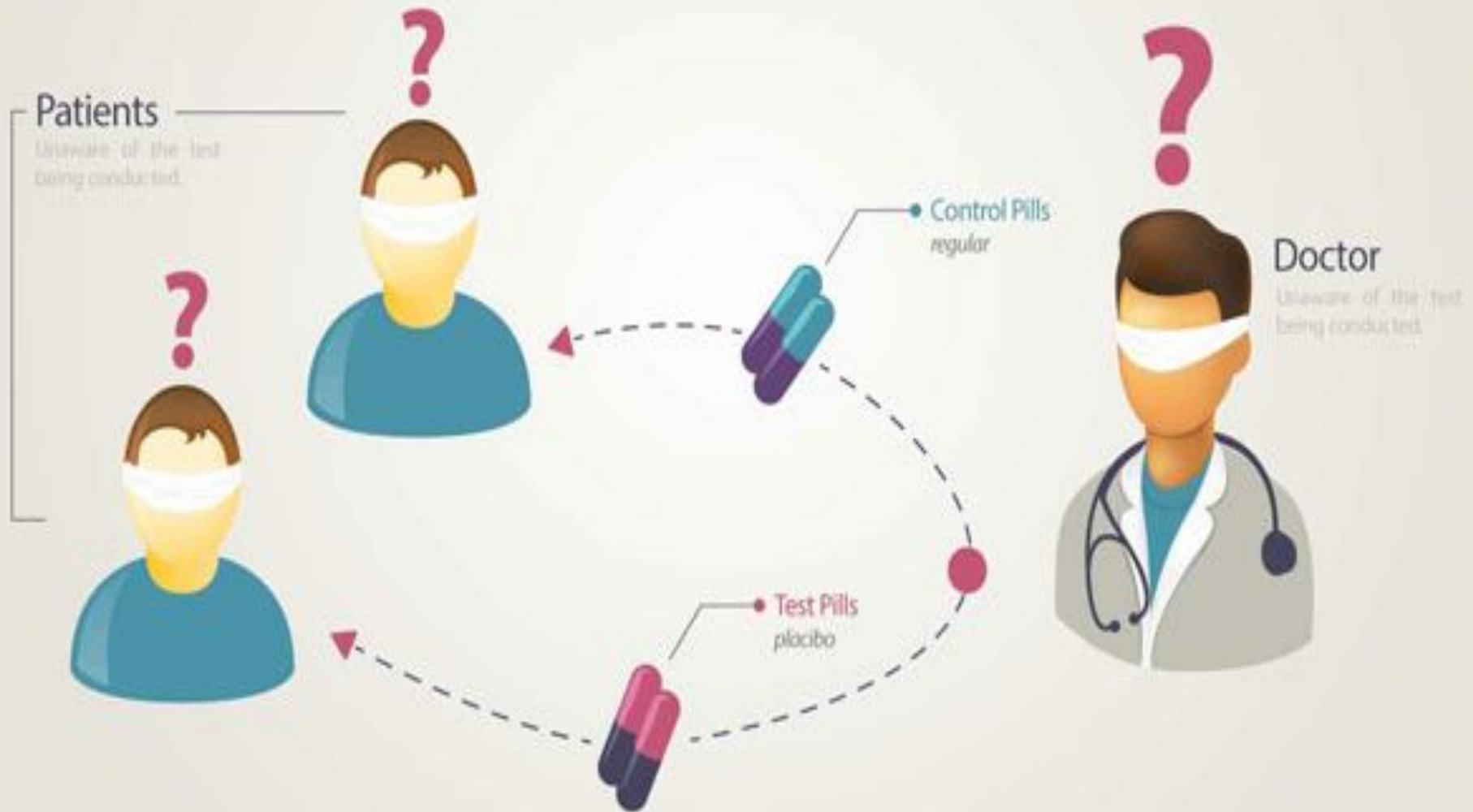


Placebo effect: Something that looks like the drug but has no medical effects



DOUBLE BLIND

Where the doctor and patient don't know whether the patients are receiving the real or placebo drug.



Activity: Complete the paragraph fill using the words to explain what happens in a trial.



review

false

real

accurate

peer

doctor

Fake

blind

Placebo

trials

results

medical

patient

clinical

journal

double

single

groups

In a clinical trial patients with the real disease are split into _____. One group receive a _____ drug called a _____. This drug is made from sugar. The other group receives the _____ drug. The _____ and the _____ don't know which type of drug they have been given. This is called a _____ trial. If the doctor did know it would be called a _____ trial.

Double blind trial help the results to be more _____

Results are published in _____ and other

Scientists _____ the work to prevent _____ claims.

Review

In a clinical trial patients with the real disease are split into groups. One group receive a Fake drug called a Placebo. This drug is made from sugar. The other group receives the Real drug. The doctor and the patient don't know which type of drug they have been given. This is called a double blind trial. If the doctor did know it would be called a single blind trial.

Double blind trials help the results to be more accurate. Results are published in medical journal and other Scientists peer review the work to prevent false claims.

LOb: Understand how new drugs are developed and tested.

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key facts



Drugs are extensively test for their **dosage, efficacy** and **toxicity**.



There is a journey with developing drugs. **Laboratory testing** (pre-clinical): testing on **cells tissues**, and **lab animals**.

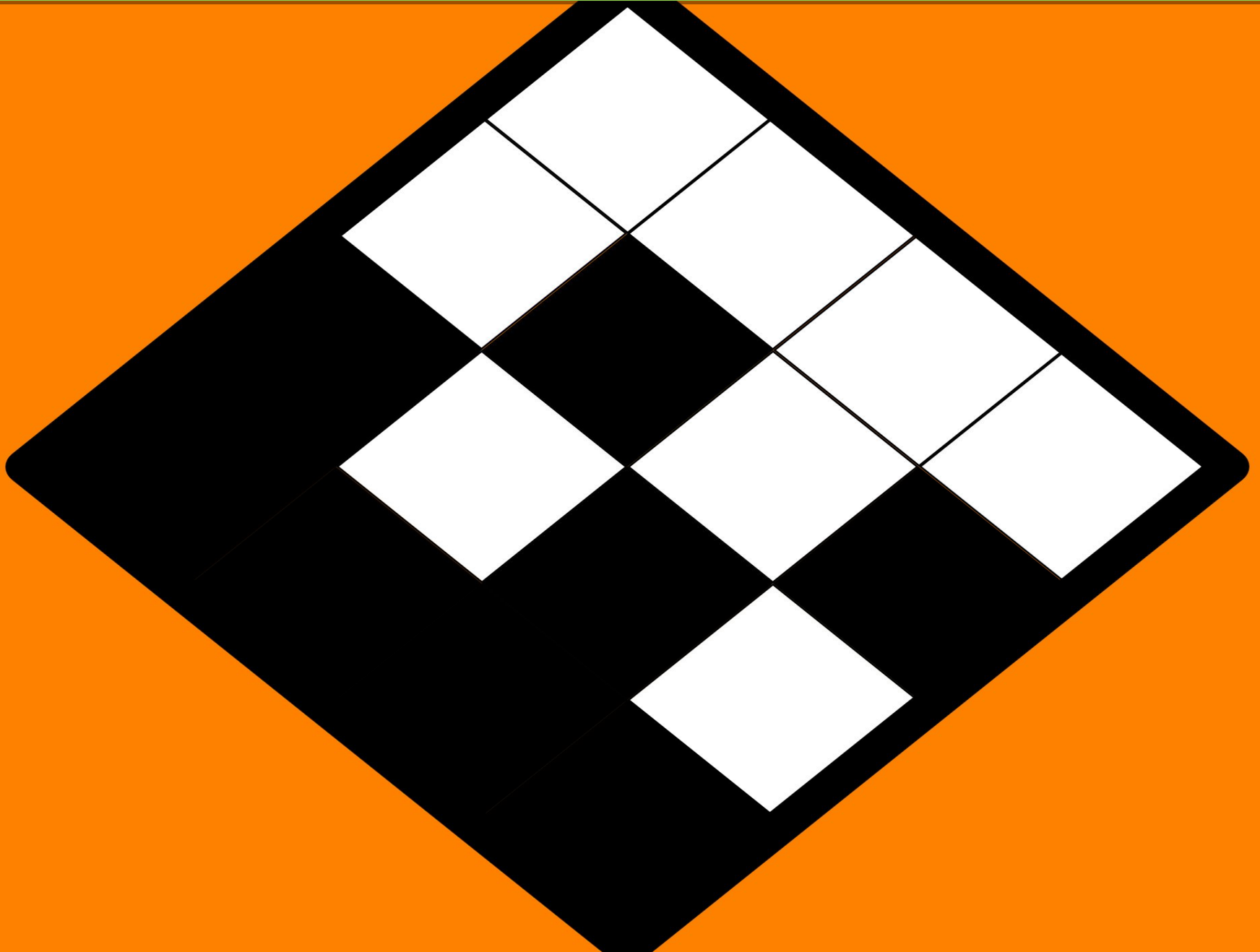


Clinical trials: stage 1: small number of healthy individuals – low dose, Stage 2: small number of patients, stage 3: large number of patients and to find optimum dose.

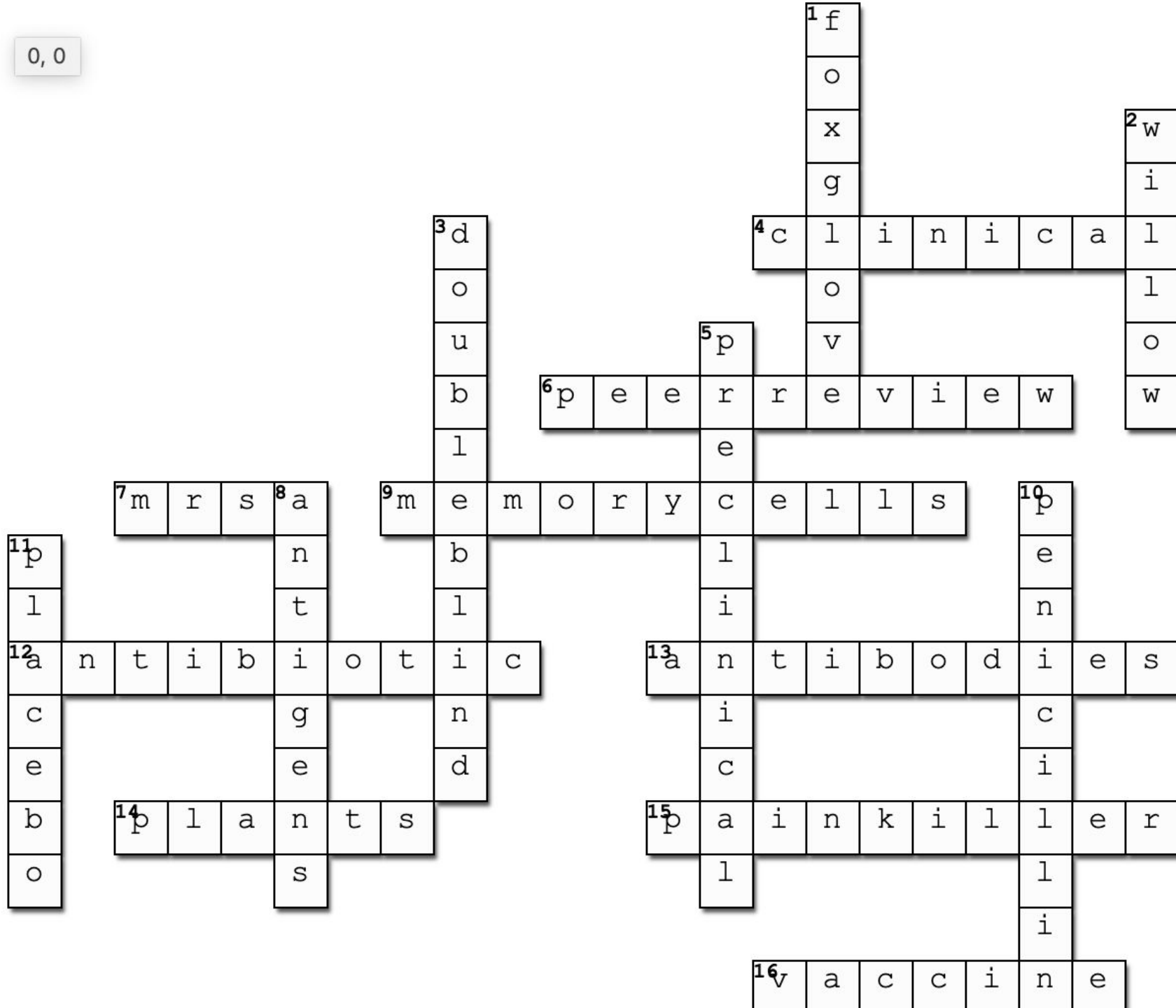


Double blind trials (patient and doctors don't know what is given to patients. **Placebo** = fake drug).

Plenary: Crossword

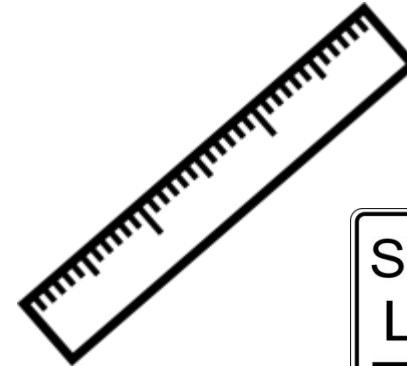
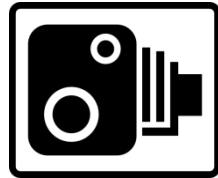


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Plenary: Triangle

Complete all the questions in this learning triangle



I am
able
to...

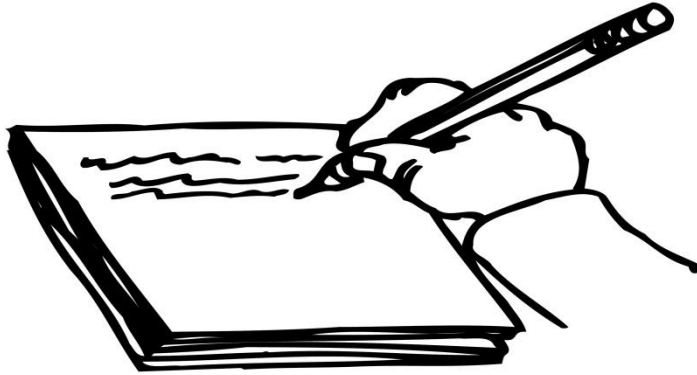


The
key Idea
I need to
understand is...

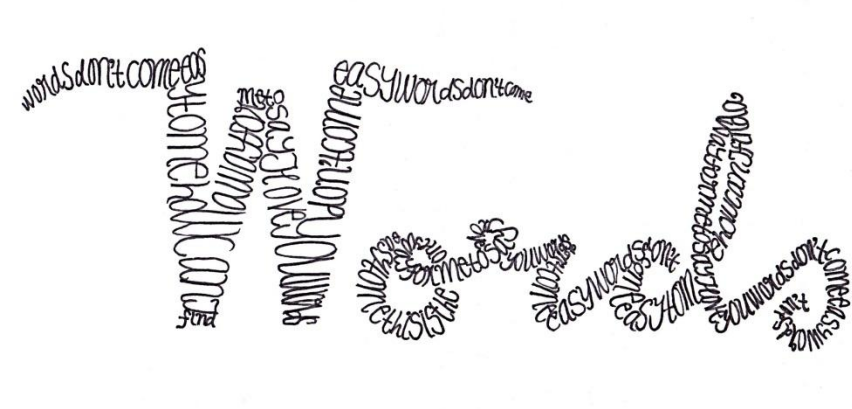
A key
word I can
explain is
_____. It
means...



Plenary: Simplify



Firstly write two sentences
to summarise your learning



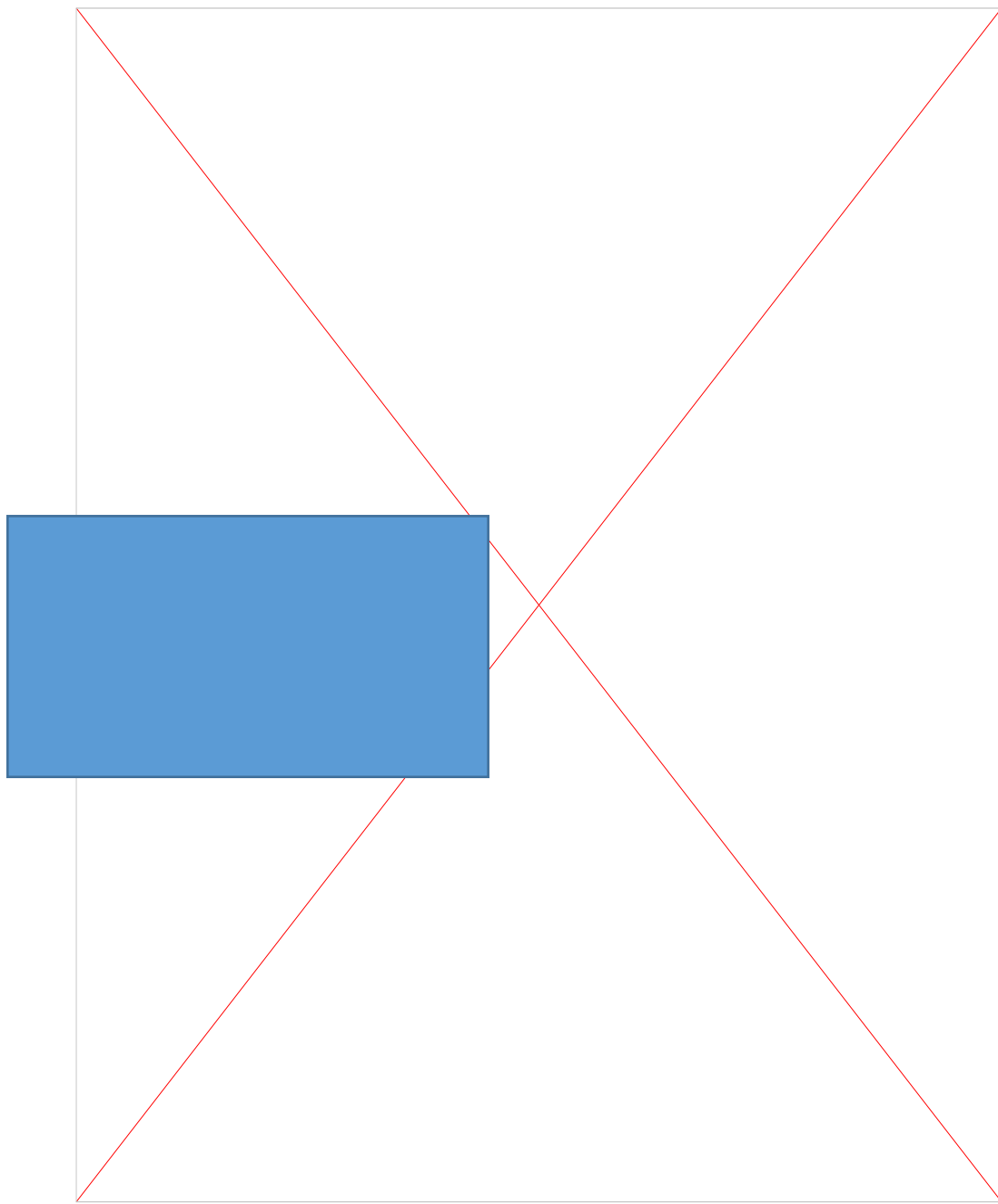
Then reduce to 5 words



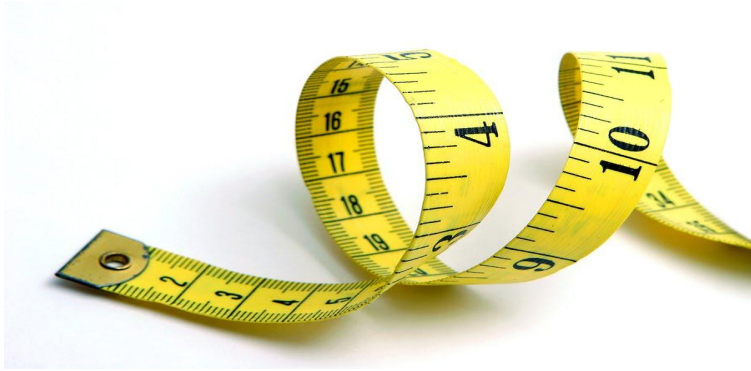
Then reduce to 1 keyword

“Investigate the memory boosting qualities of a new chocolate brand called ‘memochoc’.”





Dependent variable
What we measure



Control variable
What we keep the same



Independent variable:
What we change



Independent variable: What we change

Each person will receive a different type of chocolate:

- ☐ *Dark*
- ☐ *White*
- ☐ *None*

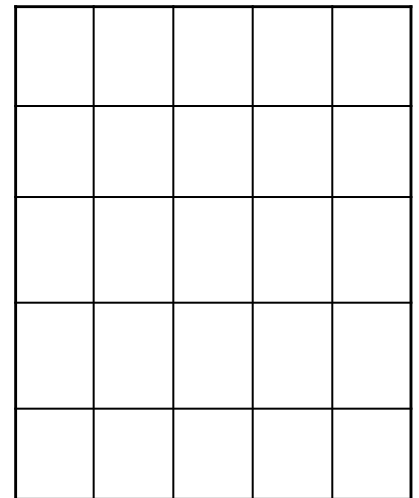
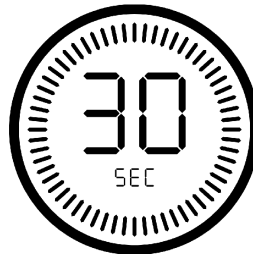


Which is memochoc? You'll see at the end.

THE TASK

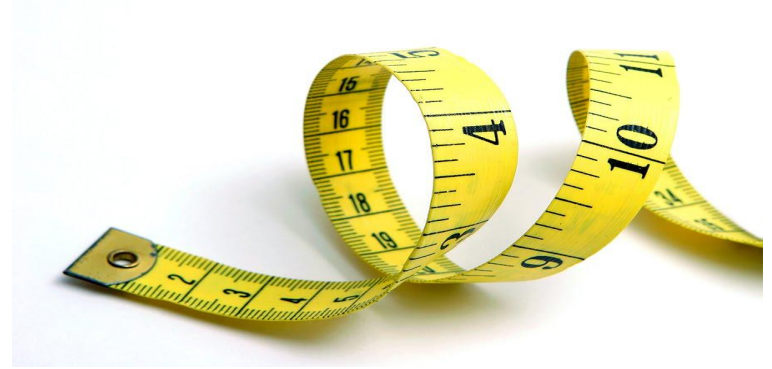
Each person will have 30s to memorise as many numbers from the grid as possible and write them onto a blank grid. They will do this first without chocolate and then after eating chocolate.

18	3	20	8	10
5	1	23	12	16
15	17	6	7	4
13	22	11	19	14
2	9	24	25	21



Dependent variable

What we measure



We will measure the **total number of squares** correctly remembered.

REMEMBER!

DON'T FORGET

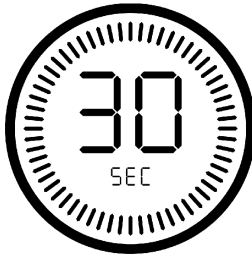
18	3			
	1			
15		6	7	4
				21

Control variable

What we keep the same



Same mass of chocolate consumed



Same amount of time

18	3	20	8	10
5	1	23	12	16
15	17	6	7	4
13	22	11	19	14
2	9	24	25	21

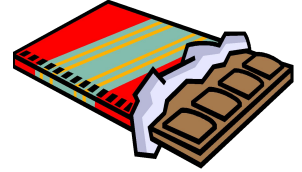
Same number grids



Groups of 4 with one scientist

1. Each person is given a memory grid by the scientist.
2. Scientist times 30s and participants remember the grid
3. Scientist gives blank grids after 30s and participants write what they remember into the correct space.
4. Participant 1 given dark chocolate. Participant 2 given white and participant 3 none.
5. Participants eat the chocolate and scientist gives 2 mins to digest
6. Repeat steps 1 – 3
7. Compare the results for your investigation.

Review questions



1. How was a **placebo** used in this investigation?
2. This trial was a **double-blind trial**. What does this mean? Why is it important to run the trial in this way?
3. Why did the creators of **#memochoc** hire the scientists to carry out the trial, rather than doing it themselves?
4. After revealing the **#** memochoc has this trial proved that it improves memory? Explain your answer.

Memory squares and blank grids

A

18	3	20	8	10
5	1	23	12	16
15	17	6	7	4
13	22	11	19	14
2	9	24	25	21

B

16	15	21	11	9
22	1	8	23	5
3	4	2	20	14
12	7	24	6	13
19	25	17	10	18

C

9	21	17	25	20
22	1	23	4	13
14	15	19	6	7
5	18	24	16	8
2	12	3	10	11

Memory squares and blank grids

A

18	3	20	8	10
5	1	23	12	16
15	17	6	7	4
13	22	11	19	14
2	9	24	25	21

B

16	15	21	11	9
22	1	8	23	5
3	4	2	20	14
12	7	24	6	13
19	25	17	10	18

C

9	21	17	25	20
22	1	23	4	13
14	15	19	6	7
5	18	24	16	8
2	12	3	10	11

